

3.4 STIR & FAT SUPPRESSION

SUPPRESSING FAT TO REVEAL PATHOLOGY

Fat is often bright on T1- and T2-weighted images and can obscure lesions, edema, and enhancement. Two main approaches are used: STIR (inversion recovery) and frequency-selective Fat Saturation (Fat Sat). Each works differently and has different strengths and failure modes.



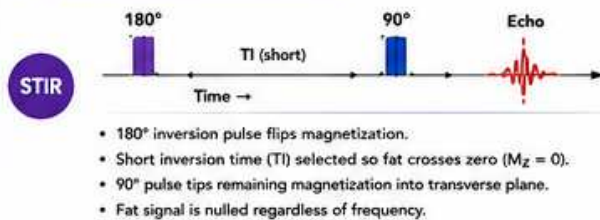
KEY CONCEPT

- STIR nulls fat by timing (short TI) → less sensitive to B_0 inhomogeneity.
- Fat Sat nulls fat by frequency → more sensitive to B_0 and center frequency.
- Choose the right technique for the patient, anatomy, and clinical goal.
- Uniform vs. localized failure gives important troubleshooting clues.

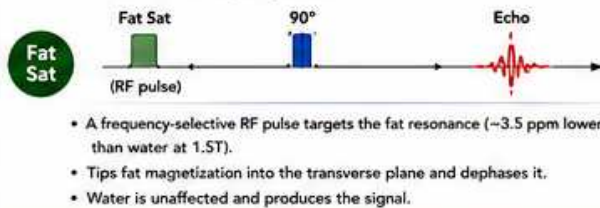
- ↑ Increase / Benefit
↓ Decrease / Trade-off
— No / Minimal Effect

1 HOW THEY WORK

A. STIR (Short Tau Inversion Recovery)



B. Fat Saturation (Frequency-Selective)



★ STIR = timing based (TI). Fat Sat = frequency based (off-resonance RF).

2 KEY DIFFERENCES

Feature	STIR	Fat Sat (Spectral)
Mechanism	Inversion recovery (short TI)	Frequency-selective RF pulse
Depends on B_0 homogeneity	Low (robust)	High (sensitive)
Depends on Center Frequency	No	Yes
Uniformity	Usually very uniform	Can be non-uniform if B_0 is off
SAR	Higher (180° pulse)	Lower
Scan Time	Slightly longer (short TI adds time)	No added time
Best Use	Difficult shimming, large FOV, implants, off-center anatomy	Post-contrast imaging, when very short TI is not desired
Common Issues	Longer TI reduces T2 contrast (lower SNR)	Local failure of fat sat (bright fat patches)

At 3T, both methods are more sensitive to B_0 inhomogeneity, but Fat Sat is still more vulnerable.

3 APPEARANCE COMPARISON (KNEE, PD-TSE)

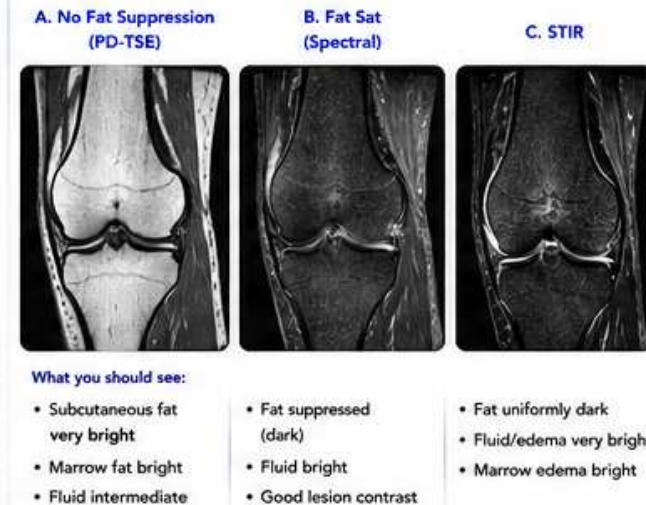


Image appearance varies with TR, TE, coil, field strength and patient. These are general expectations.

4 WHEN TO USE WHICH?

Clinical Scenario	Better Choice	Why
Post-contrast imaging (MSK, spine, body)	Fat Sat	Shorter TI preserves T1 contrast and SNR after contrast.
Edema / Inflammation (trauma, marrow edema)	STIR or Fat Sat	Both improve lesion conspicuity; STIR more robust if B_0 poor.
Large FOV, off-center anatomy	STIR	More uniform fat suppression.
Metal implants / dental hardware nearby	STIR	Less sensitive to field distortion and off-resonance.
3T with excellent shim	Fat Sat	Higher SNR and better T1 contrast after contrast.
Very short TE sequences (GRE, SSFP)	Fat Sat	STIR not available; use spectral fat sat.

★ RULE OF THUMB

If fat sat looks patchy → think B_0 / shimming / center frequency / susceptibility. If STIR is still uniform → hardware is likely OK.

5 WHAT CAN GO WRONG?

Problem / Artifact	Appearance	Likely Cause (Engineering View)	More Likely With
Incomplete Fat Sat (Local bright fat)		- Poor B_0 homogeneity (shim not optimal) - Incorrect center frequency (off resonance) - Patient off-center / large FOV - Susceptibility from metal or air/tissue interfaces	Fat Sat
Incomplete Fat Sat (Wide area)		- Global B_0 offset too large - Wrong center frequency for the field - Transmit/Receive frequency drift	Fat Sat
Too much signal loss (over-suppression)		- Fat Sat pulse too broad or power too high - Center frequency on water (not fat) - Timing issue with spectral pulse in some sequences	Fat Sat
Lower SNR with STIR		- Very short TI reduces longitudinal recovery - Longer TR / TI increases scan time	STIR
Residual fat with STIR (rare)		- TI not set to null fat (wrong TI or wrong field) B_0 errors extreme (fat null not achieved)	STIR

★ Uniform failure → global issue (shim, frequency, RF). Patchy failure → local issue (B_0 , susceptibility, off-center).

6 ENGINEER TROUBLESHOOTING APPROACH

1 Verify the sequence	Confirm STIR vs Fat Sat selected. Check TI value (STIR) or Fat Sat method.
2 Patient & positioning	Is the anatomy centered? Large FOV? Any dental work, implants, or braces?
3 Coil check	Correct coil? Fully connected? No coil element failures?
4 Prescan / Center Frequency	Repeat prescan. Ensure CF is correct (water peak at 0 ppm).
5 Shim (B_0 optimization)	Run auto/local shim. Check field map if available.
6 Re-evaluate	Reacquire and review uniformity.
7 If still failing	Check for gradient / RF issues. Escalate for advanced service.

Fat Sat problems are usually B_0 related. STIR problems are usually TI or extreme B_0 .

7 PARAMETERS THAT MATTER

A. STIR	
TI (Inversion Time)	Critical – sets fat null point
TR	Longer TR improves T2 contrast & SNR
TE	Longer TE ↑ T2 contrast (↑ blurring)
ETL	Higher ETL ↑ speed (↓ blurring)
B. FAT SAT (SPECTRAL)	
Fat Sat Pulse	Bandwidth & power affect uniformity
Center Frequency (CF)	Must be set to fat resonance
TR / TE	Same principles as the base sequence
Shim	Good B_0 homogeneity is critical

8 QUICK REFERENCE

STIR	- Robust to B_0 inhomogeneity. - Uniform fat suppression. - Uses short TI to null fat. - Slightly longer scan time / higher SAR.
Fat Sat	- Best T1 contrast after contrast. - Shorter scan time. - Sensitive to B_0 and CF errors. - Can fail locally (bright fat patches).



KEY TAKEAWAY

Use STIR when uniformity is critical or B_0 is challenging. Use Fat Sat when you need short TI and best post-contrast performance. Understand why each fails to fix it fast.